

Abdullah Almohammed

Independent Researcher

abdullah.m.almohammed@gmail.com

ORCID: 0009-0001-0832-0995

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The Editors

Artificial Intelligence and Law

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Dear Editors,

I am pleased to submit the enclosed manuscript, “Vertical Elaboration or Horizontal Integration? A Citation-Network Analysis of Saudi Arabian Legislation,” for consideration in *Artificial Intelligence and Law*.

The paper reports what is, to my knowledge, the first citation-network analysis of an Arab legal system. Network analysis of legal citation has a substantial literature for the United States and Europe, but the legal systems of the Arab world have remained outside it for want of machine-readable statutory corpora. Working from a recently released, provenance-annotated corpus of Saudi legislation, I construct and analyse a network over 290 legislative instruments and 410 validated inter-instrument citations.

Three contributions may interest the journal’s readership. First, a methodological one: citation extraction over statutory text involves both detecting a citation and resolving it to an instrument, and network studies built on automatic extraction inherit the errors of the second step silently. I validate every resolved edge, adjudicate the residue by hand, and report a 3.3% misresolution rate rather than assuming it away — a practice I argue should be standard. Second, a structural one: separating *vertical* citations, in which a subordinate instrument elaborates its own parent, from *horizontal* citations across instrument families changes the results materially. Under a naive family definition the Labour Law appears to co-lead the system with the Companies Law; once its eight annexes are correctly attributed, its horizontal citation count falls by more than a third and the Companies Law stands alone. The distinction generalises to any jurisdiction that pairs statutes with implementing regulations. Third, an applied one: intersecting the citation network with a hand-classified supersession graph makes a previously invisible defect class detectable at scale — live instruments that still cite repealed law — with direct implications for retrieval-augmented legal AI, which can otherwise ground an answer faithfully in a provision that points at law no longer in force.

The corpus, the citation and supersession graphs, and the analysis and figure scripts that reproduce every number and figure in the manuscript are openly available under the MIT licence and archived on Zenodo with a citable DOI (10.5281/zenodo.22019183).

The manuscript is original, has not been published previously, and is not under consideration elsewhere. It cites a companion paper describing the underlying corpus, which is currently under review at another journal; the two are distinct manuscripts with distinct contributions, and the present paper is self-contained. There are no competing interests and no funding to declare. I am the sole author.

Thank you for your consideration.

Yours sincerely,

Abdullah Almohammed